

Unit 5 KEY

1. Identify features of rational functions

$$a) f(x) = \frac{x-6}{x-3}$$

$$x\text{-int: } (6, 0)$$

$$y\text{-int: } (0, 2)$$

$$VA: x=3$$

$$HA: y=1$$

Holes: None

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow 1$
As $x \rightarrow -\infty$, $f(x) \rightarrow 1$

$$b) g(x) = \frac{x^2-25}{x+5}$$

$$\Rightarrow g(x) = \frac{(x+5)(x-5)}{x+5} = x-5, x \neq -5$$

$$x\text{-int: } (5, 0)$$

$$y\text{-int: } (0, -5)$$

VA: No VA

HA: No HA

Holes: $(-5, -10)$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$
As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

$$c) h(x) = \frac{x^2+x-6}{2x^2-8}$$

$$\Rightarrow h(x) = \frac{(x+3)(x-2)}{2(x+2)(x-2)}$$

$$= \frac{x+3}{2(x+2)}, x \neq 2$$

$$x\text{-int: } (-3, 0)$$

$$y\text{-int: } (0, \frac{3}{4})$$

$$VA: x=-2$$

$$HA: y = \frac{1}{2}$$

Holes: $(2, \frac{5}{8})$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \frac{1}{2}$
As $x \rightarrow -\infty$, $f(x) \rightarrow \frac{1}{2}$

$$1. \quad j(x) = \frac{x-1}{x^2-2x+1}$$

$$\Rightarrow j(x) = \frac{x-1}{(x-1)^2} = \frac{1}{x-1}, \quad x \neq 1$$

These are equal all the time

$$\frac{6x^2+9x-2x-3}{(2x+3)(3x-1)} = \frac{-18}{7}$$

x-int: None

y-int: $(0, -1)$

VA: $x=1$

HA: $y=0$

Holes: None

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow 0$
As $x \rightarrow -\infty$, $f(x) \rightarrow 0$

$$e) \quad k(x) = \frac{6x^2+7x-3}{3x^2-13x+4}$$

$$\Rightarrow k(x) = \frac{(2x+3)(3x-1)}{(x-4)(3x-1)}$$

$$= \frac{2x+3}{x-4}, \quad x \neq \frac{1}{3}$$

x-int: $(-3/2, 0)$

y-int: $(0, -3/4)$

VA: $x=4$

HA: $y=2$

$$\frac{2(\frac{1}{3})+3}{\frac{1}{3}-4} = \frac{\frac{11}{3}}{-\frac{11}{3}}$$

Holes: $(\frac{1}{3}, -1)$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow 2$
As $x \rightarrow -\infty$, $f(x) \rightarrow 2$

1. Identify the major attributes of each rational function. That is, give the x - and y -intercepts, vertical and horizontal asymptotes, removable discontinuities (holes), describe the end behavior of the function and its domain and range.

a) $f(x) = \frac{x-6}{x-3}$

b) $g(x) = \frac{x^2-25}{x+5}$

c) $h(x) = \frac{x^2+x-6}{2x^2-8}$

d) $j(x) = \frac{x-1}{x^2-2x+1}$

e) $k(x) = \frac{6x^2+7x-3}{3x^2-13x+4}$

**I put this one on here because you have to be careful factoring and work with fractions*

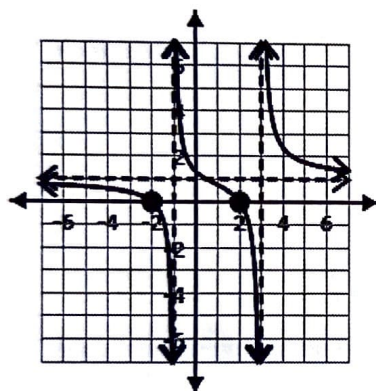
2. Create a rational function of the form $f(x) = \frac{p(x)}{q(x)}$ that satisfies the following:

- x -intercept at $(-4, 0)$
- Vertical asymptote at $x = 5$
- Hole at $x = -1$
- Horizontal asymptote at $y = \frac{3}{2}$

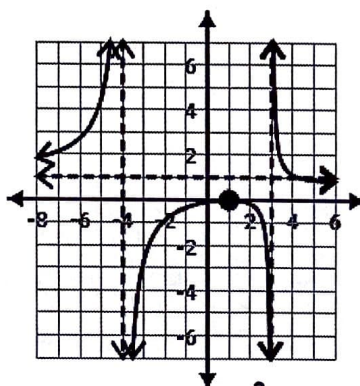
Degrees must be =

$$\frac{3(x+4)(x+1)}{2(x-5)(x+1)}$$

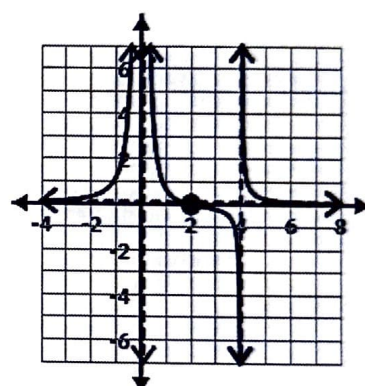
3. Write a rational function in factored form that might generate each of the graphs below.



$$f(x) = \frac{(x+2)(x-2)}{(x+1)(x-3)}$$



$$f(x) = \frac{(x-1)^2}{(x+4)(x-3)}$$



$$f(x) = \frac{x-2}{x^2(x-4)}$$

4. Evaluate the following limits given the graph of $f(x) = \frac{x^2+6x+5}{x^2-1}$.

a) $\lim_{x \rightarrow 4} f(x) = 3$

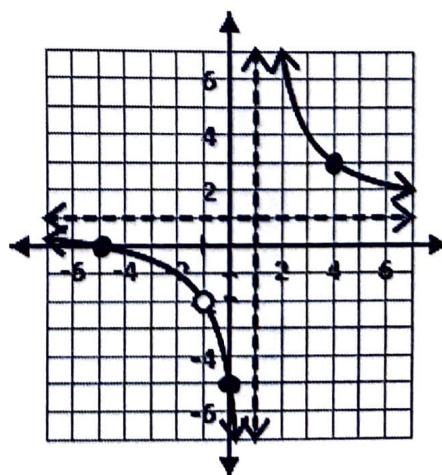
b) $\lim_{x \rightarrow 0} f(x) = -5$

c) $\lim_{x \rightarrow -1} f(x) = -2$

d) $\lim_{x \rightarrow -5} f(x) = 0$

e) $\lim_{x \rightarrow \infty} f(x) = 1$

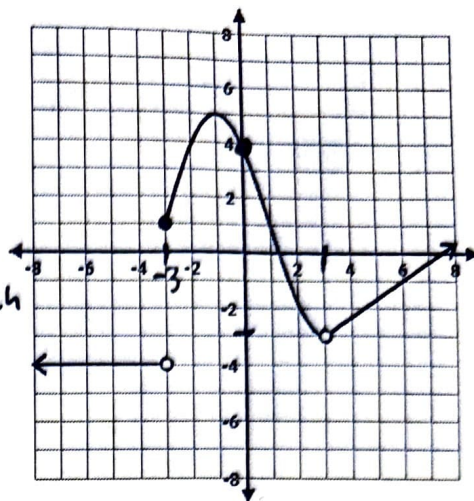
** $\lim_{x \rightarrow \infty}$ and $\lim_{x \rightarrow -\infty}$ go to the horizontal asymptote for rational functions*



5. Based off the graph of $f(x)$ shown, evaluate the following.

Write "DNE" if the limit does not exist.

- a) $\lim_{x \rightarrow 0} f(x) = 4$
- b) $\lim_{x \rightarrow \infty} f(x) = \infty$ going upwards as $x \rightarrow \infty$
- c) $\lim_{x \rightarrow -3} f(x) = \text{DNE}$ Left & Right limit don't match
- d) $\lim_{x \rightarrow 3} f(x) = -3$ Hole, yes, but both sides approach y-value at -3



6. Find the derivative of each function.

- a) $f(x) = \frac{1}{6}x^2 - 3x + 7$ $f'(x) = \frac{1}{3}x - 3$
- b) $g(x) = \sqrt[3]{x} + x$ $g'(x) = \frac{1}{3}x^{-2/3} + 1$ or $g'(x) = \frac{1}{3\sqrt[3]{x^2}} + 1$
- c) $h(x) = \frac{4}{x^2} - \frac{1}{x}$ $h'(x) = -8x^{-3} + x^{-2}$ or $h'(x) = \frac{-8}{x^3} + \frac{1}{x^2}$
- $h(x) = 4x^{-2} - x^{-1}$

7. Answer the following given $p(x) = 4\sqrt{x} + 1$.

- a) What is the average rate of change of $p(x)$ over the interval $[0, 9]$?
*Not said to be tested, still good review.

- b) What is the instantaneous rate of change of $p(x)$ at $x = 4$?

8. Answer the following given $f(x) = x^3 - 6x^2 + 5$.

- a) What is the instantaneous rate of change of $f(x)$ at $x = -2$?
- b) Over what interval is f increasing?
- c) Over what interval is f decreasing?
- d) Identify all extrema and classify each as a relative maximum or minimum value.

9. Answer the following given $f(x) = x - 2\sqrt{x}$.

- a) Over what interval is f increasing?
- b) Over what interval is f decreasing?
- c) Identify all extrema and classify each as a relative maximum or minimum value.

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$$7. f(x) = 4\sqrt{x} + 1$$

a) What is Avg Rate of Change on $[0, 9]$?

$$\frac{f(9) - f(0)}{9 - 0} = \frac{13 - 1}{9} = \frac{12}{9} = \boxed{\frac{4}{3}}$$

b) What is instantaneous RoC at $x=4$?

$$f(x) = 4x^{1/2} + 1$$

$$f'(x) = 2x^{-1/2} = \frac{2}{\sqrt{x}}$$

$$f'(4) = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1$$

$$\boxed{f'(4) = 1}$$

8. Given $f(x) = x^3 - 6x^2 + 5$.

a) Instantaneous RoC at $x = -2$

$$f'(x) = 3x^2 - 12x$$

$$\begin{aligned} f'(-2) &= 3(-2)^2 - 12(-2) \\ &= 3(4) + 24 = 12 + 24 \end{aligned}$$

$$f'(-2) = 36$$

b), c) and d)

$$\text{Set } f'(x) = 3x^2 - 12x = 0$$

$$\Rightarrow 3x(x-4) = 0$$

$$\Rightarrow x = 0, x = 4$$



$f'(-1)$ is $\oplus$ $\oplus \rightarrow \ominus$ is a max $\bigwedge$
 $f(1)$ is $\ominus$ $\ominus \rightarrow \oplus$ is a min $\bigvee$
 $f(5)$ is $\oplus$

$$f(0) = 0^3 - 6(0^2) + 5 = 5$$

$$\begin{aligned} f(4) &= 4^3 - 6(4)^2 + 5 \\ &= 64 - 96 + 5 = -27 \end{aligned}$$

b) f is increasing on $(-\infty, 0) \cup (4, \infty)$

c) f is decreasing on $(0, 4)$

d) $(0, 5)$ is a local maximum

$(4, -27)$ is a local minimum

9. Given $f(x) = x - 2\sqrt{x} = x - 2x^{1/2}$

a), b), and c)

★ Note that f 's domain is NOT $\mathbb{R}$, but limited to $x \geq 0$ since we have $\sqrt{x}$

$$f'(x) = 1 - x^{-1/2}$$

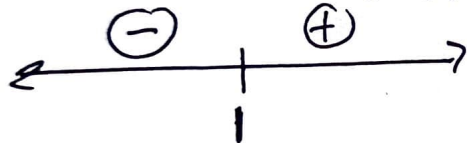
$$= 1 - \frac{1}{\sqrt{x}}$$

Set $f'(x) = 0$

$$1 - \frac{1}{\sqrt{x}} = 0$$

$$\Rightarrow 1 = \frac{1}{\sqrt{x}} \Rightarrow \sqrt{x} = 1$$

$$\therefore x = 1$$



$f'(0)$ DNE, try $f'(\frac{1}{2})$ which is $\ominus$

$f'(2)$ is $\oplus$

From $\ominus \rightarrow \oplus$ is a minimum $\checkmark$

Keep in mind f 's domain is $x \geq 0$

$$f(1) = 1 - 2\sqrt{1}$$

$$= 1 - 2$$

$$= -1$$

a) f is increasing on $(1, \infty)$

b) f is decreasing on $(0, 1)$

c) $(1, -1)$ is a local minimum